

Steroidal vs. non-steroidal mineralocorticoid receptor antagonists in heart failure with preserved ejection fraction: a propensity-matched multi-network nationwide cohort study

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Aims

Heart failure (HF) continues to portend significant morbidity and mortality within the USA. Data regarding the efficacy of mineralocorticoid receptor antagonists (MRAs) in HF with preserved ejection fraction (HFpEF) remains unclear. Finerenone, a non-steroidal MRA, has demonstrated reduction in hospitalizations in this population. However, the comparative efficacy of different MRA classes in these patients remains elusive.

Methods and results

We conducted a retrospective observational cohort study using the TriNetX research network. We applied 1:1 propensity score matching (PSM) to compare patients with HFpEF initiated on finerenone vs. spironolactone. We examined 1-year risks of HF exacerbation, acute kidney injury (AKI), electrolyte derangements, and cardiovascular outcomes. We identified 254,417 patients with HFpEF and MRA use. After PSM, each cohort contained 491 patients. Analysis revealed that finerenone use was associated with a significantly lower risk of HF exacerbation (hazard ratio [HR] = 0.63 [95% confidence interval [CI]: 0.54, 0.74], $P < 0.001$), all-cause mortality (HR = 0.38 [95% CI: 0.23, 0.64], $P < 0.001$), and hypokalaemia (HR = 0.58 [95% CI: 0.40, 0.85], $P = 0.001$). The aforementioned results, along with a reduction in AKI risk, were statistically significant after stratified subgroup analysis by diabetes mellitus and chronic kidney disease.

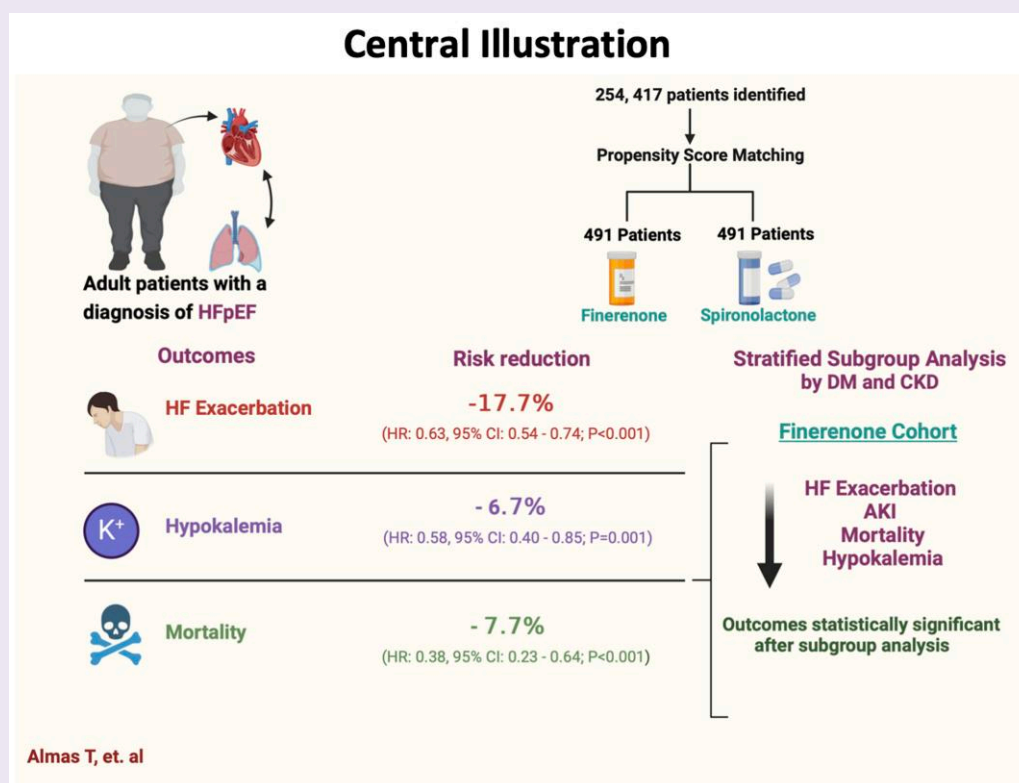
Conclusion

In this large real-world HFpEF cohort, finerenone use was associated with lower risks of HF exacerbation and all-cause mortality compared with spironolactone. These observational findings are hypothesis-generating and warrant confirmation in randomized trials.

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Graphical abstract



Keywords

Heart failure • Heart failure with preserved ejection fraction • Mineralocorticoid receptor antagonists • Hyperkalemia • Morbidity • Spironolactone • Finerenone

Key Learning Points

What is already known:

- Mineralocorticoid receptor antagonists (MRAs) have uncertain efficacy in heart failure with preserved ejection fraction (HFpEF), with limited comparative data between MRA classes.
- Finerenone, a non-steroidal MRA, has previously demonstrated a reduction in hospitalizations in HFpEF patients.
- HFpEF continues to have significant morbidity and mortality with limited therapeutic options.

What this study adds:

- Real-world comparative analysis showing finerenone is associated with significantly lower risk of HF exacerbations compared to spironolactone in patients with HFpEF.
- Finerenone demonstrated a favourable side effect profile with reduced hyperkalaemia risk compared to spironolactone.

Introduction

Community-based epidemiological studies report that roughly half of all heart failure (HF) patients are classified as HF with preserved ejection fraction (HFpEF).¹ Despite its high morbidity, mortality, and ever-increasing healthcare expenditures, evidence-based pharmaceutical interventions for HFpEF remain

limited.²⁻⁴ Maladaptive renin-angiotensin-aldosterone system (RAAS) activation remains a cornerstone of the disease's pathophysiology, driving sodium retention alongside myocardial fibrosis and subsequent cardiac remodelling.⁵ Consequently, mineralocorticoid receptor antagonists (MRAs) have been investigated as an option to blunt the RAAS cascade and potentially improve HFpEF outcomes.

Steroidal MRAs like spironolactone and eplerenone have shown clear mortality benefits in patients with HF with reduced ejection fraction (HFrEF), as exemplified by the Randomized Aldactone Evaluation Study and Eplerenone in Mild Patients Hospitalization and Survival Study in Heart Failure trials.^{6,7} Yet, TOPCAT, the largest randomized trial studying spironolactone in HFpEF, was met with equivocal results for the primary composite outcome of cardiovascular mortality and HF hospitalizations.⁸ However, geographic heterogeneity in patient phenotype, treatment compliance, and event rates complicates the study's interpretation.⁹ When combined with concerns of hyperkalaemia and worsening renal function, clinicians are left uncertain about spironolactone's role in HFpEF.¹⁰

Finerenone, a novel non-steroidal and selective MRA, has demonstrated promise to overcome the drawbacks seen with steroidal MRAs while maintaining their clinical benefits.^{11,12} In Phase II studies, finerenone was associated with lower rates of hyperkalaemia and deterioration of kidney function when compared with spironolactone and eplerenone in patients with chronic kidney disease (CKD).^{13,14} Recently, the FINEARTS-HF trial investigated finerenone within the HFpEF patient population, where finerenone significantly reduced total HF hospitalization with a tolerable side effect profile.¹⁵ While the reduction in cardiovascular death was not statistically significant, the trial's findings renewed interest in MRAs as a potential option in HFpEF's limited therapeutic arsenal.

To date, no head-to-head randomized controlled trial has directly compared finerenone to spironolactone in HFpEF patients. Indirect comparisons of separate trials have their limitations as they inevitably include heterogeneous patient populations and different outcome measures. In addition, real-world efficacy and safety profiles often differ from those observed in strictly controlled trials, necessitating the need for complementary observational studies to help guide pragmatic recommendations.¹⁶

Large healthcare databases, such as TriNetX or Datavant, help fill this gap by facilitating robust retrospective analyses to approximate randomized comparisons across diverse patient populations and healthcare settings. To this end, we performed a propensity-matched retrospective cohort study using the TriNetX database to compare finerenone and spironolactone in patients with HFpEF. By building upon the foundations established by the FINEARTS-HF trial, our study aims to leverage real-world data to elucidate the differences in clinical outcomes, including hospitalization rates, mortality, and adverse events, thereby addressing a key knowledge gap in HFpEF management.

Methods

Patient and data selection

We performed a retrospective observational cohort study using the TriNetX database. TriNetX is a de-identified federated database with 105 contributing healthcare organizations (HCOs), including over 140 million patients (TriNetX Research Network, Cambridge, MA; date of data access: Feb 20, 2025). TriNetX continuously aggregates clinical data directly from the electronic medical records of participating HCOs and performs extensive data quality and accuracy assessment. While TriNetX obfuscates institutional information relating to participating HCOs, a typical HCO includes a large academic health centre with inpatient, outpatient, and specialty care services. The database reports patient and outcome data in an anonymized, de-identified manner, and as such, the present study was exempt from Institutional Review Board approval.

Using the International Classification of Disease 10th edition (ICD-10), we identified all patients with a diagnosis of HFpEF (identified using ICD-10 code I50.3 recorded in the electronic health

record). Additional data points to support a diagnosis of HFpEF, such as echocardiographic parameters and New York Heart Association (NYHA) functional class, are not available consistently within the TriNetX database, limiting the ability to incorporate these into our definition of HFpEF. Furthermore, we acknowledge the potential misclassification of patients within the HFpEF category, with the possibility of those with HF with improved or mid-range ejection fraction being labelled as pEF. We further stratified these patients into two distinct cohorts based on finerenone or spironolactone use within 1-year of HFpEF diagnosis (Finerenone, RxNorm: 2562811562811; Spironolactone, RxNorm: 9997). Currently, finerenone does not have a regulatory indication for HFpEF; in our cohort, finerenone was most likely prescribed for its approved indications in patients with CKD and/or type 2 diabetes who also carried a diagnosis of HFpEF. Imperatively, when defining cohorts, particular caution was taken to exclude concomitant spironolactone and/or finerenone use in either cohort. Time zero (TO) was defined as the date of the first recorded prescription for finerenone or spironolactone after the HFpEF diagnosis. Only events occurring after this index date were counted towards the 1-year outcomes; pre-index events were not included in the outcome assessment. For both cohorts, we collected data on patient demographics, baseline characteristics, background medication use, and comorbidities. This analysis included outcomes that occurred in the time window that started 1 day after the first occurrence of the index event (finerenone or spironolactone initiation) and ended 365 days, or 1 year, after the first occurrence of the index event. Patients whose index event occurred 20 years or more ago were excluded from the analysis. NYHA functional class data were not available in structured fields within TriNetX and could not be incorporated into our analysis. Comprehensive details of the codes used to define cohorts are detailed in the [Supplementary material online, Appendices SA and SB](#). The present observational study was in concordance with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines for reporting observational studies.

Study outcomes and statistical analysis

The outcomes studied in the present analysis included first or subsequent HF exacerbations (pulmonary oedema [ICD-10: J81] or loop diuretic, metolazone, or chlorothiazide use), AKI (ICD-10: N17), renal replacement therapy (CPT group: 1012740), acute myocardial infarction (AMI, ICD-10: I21), ischaemic stroke (ICD-10: I63), all-cause mortality, sudden cardiac death (SCD, ICD-10: I49.01), hypokalaemia (ICD-10: E87.6), and hyperkalaemia (ICD-10: E87.5) within a 1-year follow-up period. AMI and ischaemic stroke were included as secondary exploratory cardiovascular outcomes, with the aim of highlighting the elevated vascular risk profile of patients with HFpEF and the broader cardiometabolic context in which these MRAs are used, and are not HF-specific endpoints. Additionally, we performed separate sub-analyses within patients with the comorbid condition of CKD (ICD-10: N18), and/or diabetes mellitus (DM, ICD-10: E08-E11). The rationale for using thiazides or metolazone as part of our definition for HF exacerbation is that in refractory cases, sequential diuresis with these agents is used in an inpatient setting to augment diuresis, especially in patients with higher diuretic thresholds. Therefore, along with intravenous diuretic use, thiazide/metolazone formed part of our definition for HF exacerbation.

To mitigate any existing confounders, we performed 1:1 greedy nearest-neighbour PSM without replacement, applying a calibre of 0.1 on the logit of the propensity score based on age, ethnicity, gender, baseline comorbidities, and medication use ([Table 1](#)). We performed independent *t*-tests for continuous data and chi-square tests for categorical data (presented as frequencies and percentages), where applicable. We evaluated differences in outcome risk using hazard ratios (HRs) and 95% confidence intervals (CIs). This analytical framework was repeated within patients with CKD and/or DM subsets. Pertinently, we propensity-matched the two cohorts based on the criteria listed within the Charleston comorbidity index, including age, ethnicity, gender, baseline comorbidities, and medication use, including the use of guideline-directed medical

Table 1 Baseline characteristics for patients with HFpEF with finerenone or spironolactone before and after propensity score matching

	Before matching			After matching		
	Finerenone	Spironolactone	P-value	Finerenone	Spironolactone	P-value
<i>n</i>	497	253 920		491	491	
Age at index	70.26 ± 11.26	69.49 ± 14.05	0.222	70.25 ± 11.3	69.68 ± 12.37	0.446
Sex						
Female	206 (41.4%)	134 402 (52.9%)	<0.001	205 (41.8%)	198 (40.3%)	0.650
Male	285 (57.3%)	111 986 (44.1%)	<0.001	280 (57.0%)	285 (58.0%)	0.747
Race						
White	263 (52.9%)	173 289 (68.2%)	<0.001	263 (53.6%)	272 (55.4%)	0.564
Black or African American	75 (15.1%)	47 828 (18.8%)	0.033	75 (15.3%)	73 (14.9%)	0.858
Asian	89 (17.9%)	5617 (2.2%)	<0.001	83 (16.9%)	76 (15.5%)	0.544
American Indian or Alaska Native	10 (2.0%)	707 (0.3%)	<0.001	10 (2.0%)	10 (2.0%)	1.000
Native Hawaiian or Other Pacific Islander	14 (2.8%)	1358 (0.5%)	<0.001	14 (2.9%)	12 (2.4%)	0.691
Other race	27 (5.4%)	6039 (2.4%)	<0.001	27 (5.5%)	27 (5.5%)	1.000
Unknown Race	27 (5.4%)	19 082 (7.5%)	0.078	27 (5.5%)	29 (5.9%)	0.783
Ethnicity						
Not Hispanic or Latino	400 (80.5%)	195 618 (77.0%)	0.068	394 (80.2%)	391 (79.6%)	0.811
Diagnoses						
Atrial fibrillation and flutter	192 (38.6%)	126 642 (49.9%)	<0.001	191 (38.9%)	196 (39.9%)	0.744
Chronic kidney disease (CKD)	409 (82.3%)	104 876 (41.3%)	<0.001	403 (82.1%)	411 (83.7%)	0.498
Diabetes mellitus	453 (91.1%)	127 636 (50.3%)	<0.001	447 (91.0%)	447 (91.0%)	1.000
Essential (primary) hypertension	448 (90.1%)	210 433 (82.9%)	<0.001	443 (90.2%)	446 (90.8%)	0.744
Obstructive sleep apnea (adult) (pediatric)	213 (42.9%)	86 379 (34.0%)	<0.001	211 (43.0%)	216 (44.0%)	0.748
Old myocardial infarction	103 (20.7%)	46 992 (18.5%)	0.203	102 (20.8%)	110 (22.4%)	0.535
Overweight and obesity	295 (59.4%)	127 128 (50.1%)	<0.001	293 (59.7%)	292 (59.5%)	0.948
Medications						
Ace inhibitors	231 (46.5%)	123 220 (48.5%)	0.361	228 (46.4%)	241 (49.1%)	0.406
Angiotensin ii inhibitor	321 (64.6%)	111 552 (43.9%)	<0.001	316 (64.4%)	319 (65.0%)	0.841
Beta blockers/related	436 (87.7%)	217 822 (85.8%)	0.215	431 (87.8%)	431 (87.8%)	1.000
Glucagon-like peptide-1 (GLP-1) analogues	199 (40.0%)	16 030 (6.3%)	<0.001	193 (39.3%)	194 (39.5%)	0.948
Loop diuretics	410 (82.5%)	228 319 (89.9%)	<0.001	409 (83.3%)	420 (85.5%)	0.333
Sacubitril	46 (9.3%)	20 985 (8.3%)	0.423	46 (9.4%)	52 (10.6%)	0.523
Sodium-glucose co-transporter 2 (SGLT2) inhibitors	267 (53.7%)	38 101 (15.0%)	<0.001	261 (53.2%)	280 (57.0%)	0.223
Thiazides/related diuretics	266 (53.5%)	113 965 (44.9%)	<0.001	264 (53.8%)	293 (59.7%)	0.062
Valsartan	112 (22.5%)	46 368 (18.3%)	0.014	112 (22.8%)	120 (24.4%)	0.548

Significant P-values in bold.

therapy (GDMT) for HF. For all outcomes, wherever applicable, Kaplan–Meier survival analysis was performed by employing the log-rank test, Cox hazard ratio, and proportionality. All tests were two-tailed with an alpha level of 0.05.

Due to concerns regarding potential residual confounding and endpoint specificity, we conducted additional sensitivity analyses. First, we performed a sensitivity analysis using a more stringent definition of HF exacerbation restricted to HF-related hospitalization or emergency department visits with a primary or secondary diagnosis of HF or pulmonary oedema and concomitant loop diuretic use. Next, we performed quantitative bias analyses using *E*-values for our key outcomes (HF exacerbation, all-cause mortality, and secondary outcomes) to estimate the minimum strength of association that an unmeasured confounder would need to have with both

treatment and outcome to fully explain away the observed associations. These sensitivity analyses are presented in detail in the [Supplementary material online, Appendix](#), and are discussed within the Results section in greater detail.

Results

The present study included 254 417 patients with a diagnosis of HFpEF who were initiated on either spironolactone or finerenone within 1 year of the index diagnosis. Prior to PSM, there were 253 920 patients in the spironolactone cohort and 497 patients in the finerenone cohort. After PSM, there were 491 patients in either cohort, yielding 982 patients overall.

Table 2 Outcomes between patients with HFpEF with finerenone vs. spironolactone, after propensity score matching, with hazard ratios and 95% confidence intervals

	Finerenone	Spironolactone	HR (95% CI)	P-value
HF Exacerbation	271 (55.2%)	358 (72.9%)	0.63 (0.54–0.74)	<0.001
AKI	139 (28.3%)	177 (36.0%)	0.81 (0.65–1.02)	0.009
Renal Replacement therapy	11 (2.2%)	14 (2.9%)	0.86 (0.39–1.88)	0.543
AMI	44 (9.0%)	47 (9.6%)	1.01 (0.67–1.53)	0.741
Ischemic stroke	29 (5.9%)	41 (8.4%)	0.77 (0.48–1.23)	0.137
All-cause mortality	20 (4.1%)	58 (11.8%)	0.38 (0.23–0.64)	<0.001
Sudden cardiac death	10 (2.0%)	~1 (0.2%)	—	0.002
Hypokalaemia	41 (8.4%)	74 (15.1%)	0.58 (0.40–0.85)	0.001
Hyperkalaemia	68 (13.8%)	79 (16.1%)	0.96 (0.69–1.33)	0.325
Atrial fibrillation/flutter	151 (30.8%)	174 (35.4%)	0.92 (0.74–1.14)	0.119

Significant P-values in bold.

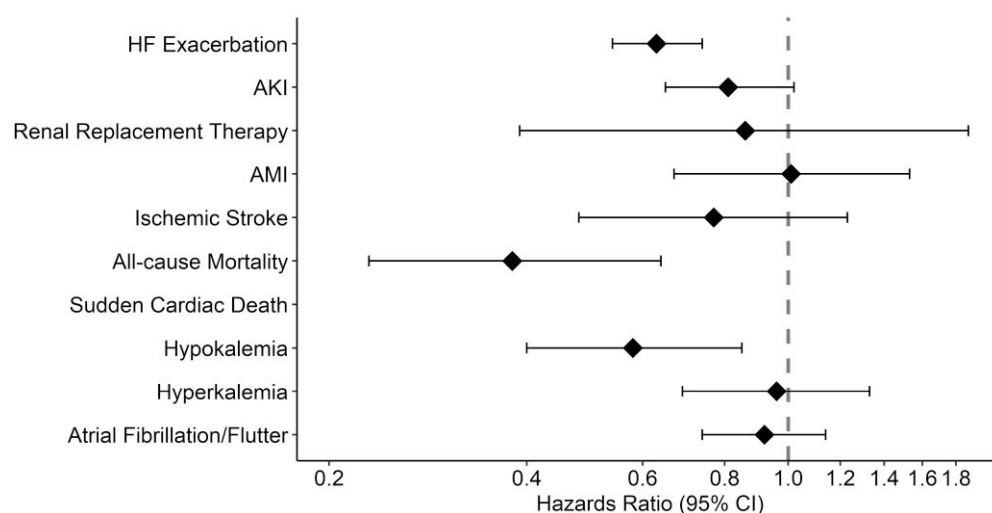
**Figure 1** Forest plot depicting the hazard ratios (HRs, diamonds) and 95% confidence intervals (CI, horizontal error bars) between finerenone use and spironolactone use (reference category) on the study outcomes (y-axis). Vertical dashed line indicates the null hypothesis (HR = 1).

Table 1 tabulates the patients' baseline demographic data, baseline medication use, and comorbidities before and after PSM.

Overall cohort outcomes

During the 1-year follow-up period employed in this study, finerenone use was associated with a significantly lower statistically significant risk of HF exacerbation compared with the spironolactone cohort after PSM [55.2% vs. 72.9%, HR: 0.63, 95% CI: 0.54–0.74, $P < 0.001$]. Similarly, finerenone use was associated with a significantly lower risk of all-cause mortality compared with spironolactone (4.1% vs. 11.8%, HR: 0.38, 95% CI: 0.23–0.64, $P < 0.001$) at a statistically significant level. When compared to spironolactone, finerenone was also associated with a lower incidence of AKI (28.3% vs. 36%, HR 0.81, 95% CI: 0.65–1.02), albeit not statistically significant. The risk of hypokalaemia was noted to be lower in the finerenone cohort vs. the spironolactone

cohort (8.4% vs. 15.1%, HR: 0.58, 95% CI: 0.40–0.85, $P = 0.001$). Although non-significantly, numerically similar trends were observed for hyperkalaemia, the risks of which were greater in the spironolactone cohort compared to the finerenone cohort (16.1% vs. 13.8%, HR: 0.96, 95% CI: 0.69–1.33). Incidence of SCD was rare in both groups, with only 1 subject within the spironolactone cohort and 10 subjects within the finerenone cohorts. Provided the quite low number of SCD events within both cohorts, and wide confidence intervals, our study was not adequately powered to detect differences in SCD, hence these findings should be interpreted with due caution.

These outcomes are delineated in **Table 2** and are depicted in **Figure 1**.

Subgroup analysis

Given the nature of the drugs of interest and their inherent role in a myriad of disease processes, we conducted a subgroup analysis to

Table 3 Baseline characteristics for patients with HFpEF with finerenone or spironolactone, stratified by CKD, before and after propensity score matching

	Before matching			After matching		
	Finerenone	Spironolactone	P-value	Finerenone	Spironolactone	P-value
<i>n</i>	415	129 117		409	409	
Age at index	70.28 ± 11.55	71.4 ± 12.95	0.080	70.22 ± 11.57	69.71 ± 11.11	0.522
Sex						
Female	167 (40.2%)	66 480 (51.5%)	<0.001	166 (40.6%)	178 (43.5%)	0.395
Male	243 (58.6%)	58 750 (45.5%)	<0.001	238 (58.2%)	229 (56.0%)	0.525
Race						
White	224 (54.0%)	85 474 (66.2%)	<0.001	224 (54.8%)	219 (53.5%)	0.726
Black or African American	65 (15.7%)	27 464 (21.3%)	0.005	65 (15.9%)	76 (18.6%)	0.309
Asian	76 (18.3%)	3064 (2.4%)	<0.001	70 (17.1%)	69 (16.9%)	0.926
American Indian or Alaska Native	10 (2.4%)	316 (0.2%)	<0.001	10 (2.4%)	10 (2.4%)	1.000
Native Hawaiian or Other Pacific Islander	12 (2.9%)	790 (0.6%)	<0.001	12 (2.9%)	12 (2.9%)	1.000
Other race	17 (4.1%)	2694 (2.1%)	0.004	17 (4.2%)	20 (4.9%)	0.614
Unknown Race	19 (4.6%)	9315 (7.2%)	0.038	19 (4.6%)	11 (2.7%)	0.137
Ethnicity						
Not Hispanic or Latino	335 (80.7%)	100 735 (78.0%)	0.184	329 (80.4%)	327 (80.0%)	0.861
Diagnoses						
Atrial fibrillation and flutter	166 (40.0%)	72 007 (55.8%)	<0.001	165 (40.3%)	172 (42.1%)	0.619
Chronic kidney disease (CKD)	415 (100.0%)	129 117 (100.0%)	<0.001	409 (100.0%)	409 (100.0%)	0.562
Diabetes mellitus	396 (95.4%)	79 484 (61.6%)	<0.001	390 (95.4%)	392 (95.8%)	0.733
Essential (primary) hypertension	375 (90.4%)	113 985 (88.3%)	0.188	370 (90.5%)	376 (91.9%)	0.459
Obstructive sleep apnea (adult) (pediatric)	180 (43.4%)	49 819 (38.6%)	0.045	179 (43.8%)	193 (47.2%)	0.326
Old myocardial infarction	92 (22.2%)	30 385 (23.5%)	0.513	92 (22.5%)	90 (22.0%)	0.866
Overweight and obesity	254 (61.2%)	72 102 (55.8%)	0.028	252 (61.6%)	254 (62.1%)	0.886
Medications						
Ace inhibitors	197 (47.5%)	68 238 (52.9%)	0.028	194 (47.4%)	212 (51.8%)	0.208
Angiotensin ii inhibitor	268 (64.6%)	62 689 (48.6%)	<0.001	263 (64.3%)	259 (63.3%)	0.771
Beta blockers/related	368 (88.7%)	117 198 (90.8%)	0.141	363 (88.8%)	353 (86.3%)	0.290
Glucagon-like peptide-1 (GLP-1) analogues	168 (40.5%)	10 302 (8.0%)	<0.001	162 (39.6%)	172 (42.1%)	0.477
Loop diuretics	355 (85.5%)	121 476 (94.1%)	<0.001	354 (86.6%)	352 (86.1%)	0.839
Sacubitril	31 (7.5%)	12 015 (9.3%)	0.199	31 (7.6%)	29 (7.1%)	0.789
Sodium-glucose co-transporter 2 (SGLT2) inhibitors	222 (53.5%)	21 778 (16.9%)	<0.001	216 (52.8%)	211 (51.6%)	0.726
Thiazides/related diuretics	225 (54.2%)	68 679 (53.2%)	0.676	223 (54.5%)	233 (57.0%)	0.481
Valsartan	85 (20.5%)	26 829 (20.8%)	0.882	85 (20.8%)	80 (19.6%)	0.663

Significant P-values in bold.

see whether the findings observed in the general cohort resonated in the subgroups. We stratified the cohort based on (A) whether or not patients had a diagnosis of CKD and (B) whether or not patients had a diagnosis of DM. As can be noted from [Table 3](#), the overall population spread did not vary after stratification.

Chronic kidney disease subgroup analysis

After stratifying patients based on their CKD status, we observed similarly favourable outcome associations within the finerenone cohort. Finerenone use was associated with significantly lower risks of HF exacerbation compared with spironolactone (59.4% vs. 74.1%, HR: 0.63, 95% CI: 0.53–0.74, $P < 0.001$). Similarly, finerenone use was associated with

significantly lower risks of AKI, hypokalaemia, and all-cause mortality compared with spironolactone. These comparative outcomes are delineated in [Table 4](#) below.

We further stratified patients by their DM status. Baseline demographics after stratification are tabulated in [Table 5](#).

Diabetes mellitus subgroup analysis

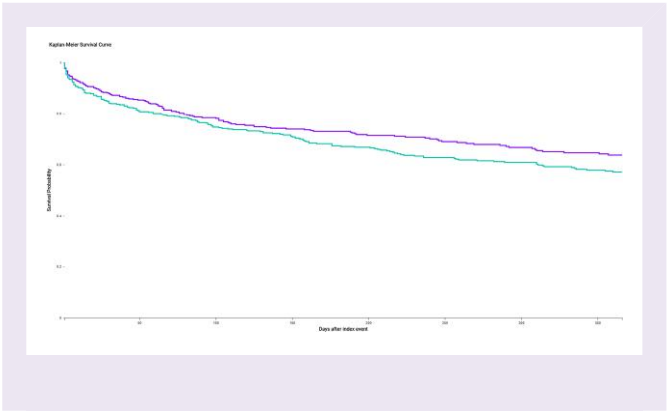
After stratifying patients with regards to their DM status, we found that finerenone use was associated with significantly lower risks of HF exacerbation compared with spironolactone (56.6% vs. 74%, HR 0.59, 95% CI: 0.50–0.69, $P < 0.001$). Similarly, finerenone use was associated with lower risks of developing an AKI, all-cause mortality, and hypokalaemia

Table 4 Outcomes between patients with HFpEF with finerenone or spironolactone, stratified by CKD, after propensity score matching, with hazard ratios and 95% confidence intervals

	Finerenone	Spironolactone	HR (95% CI)	P-value
HF exacerbation	243 (59.4%)	303 (74.1%)	0.63 (0.53–0.74)	<0.001
AKI	135 (33.0%)	168 (41.1%)	0.78 (0.62–0.98)	0.017
Renal Replacement therapy	11 (2.7%)	15 (3.7%)	0.75 (0.34–1.62)	0.425
AMI	43 (10.5%)	52 (12.7%)	0.85 (0.56–1.26)	0.326
Ischaemic stroke	25 (6.1%)	33 (8.1%)	0.77 (0.46–1.30)	0.276
All-cause mortality	19 (4.6%)	51 (12.5%)	0.38(0.23–0.65)	<0.001
Sudden cardiac death	10 (2.4%)	10 (2.4%)	0.34 (0.07–1.66)	1.000
Hypokalaemia	38 (9.3%)	67 (16.4%)	0.58 (0.39–0.86)	0.002
Hyperkalaemia	65 (15.9%)	84 (20.5%)	0.79 (0.57–1.09)	0.085
Atrial fibrillation/flutter	132 (32.3%)	153 (37.4%)	0.84 (0.67–1.06)	0.123

Significant P-values in bold.

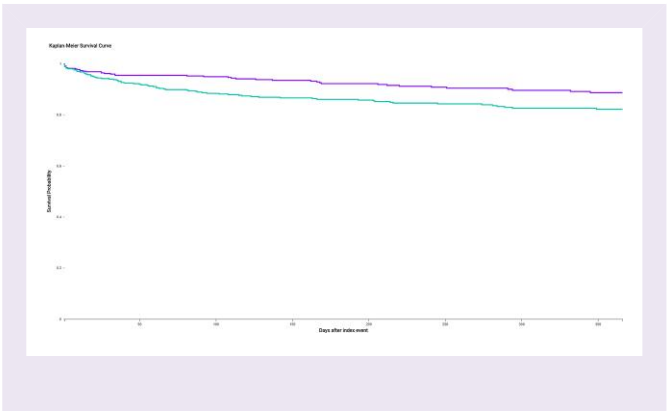
compared with spironolactone. These outcomes are elucidated in [Table 6](#).



Specific outcomes

Heart failure exacerbation

Finerenone use was associated with a significantly lower risk of HF exacerbation compared with the spironolactone group



(55.2% vs. 72.9%) with an absolute risk reduction of 17.7% (HR: 0.63, 95% CI: 0.54–0.74), with a P-value <0.001. Survival analysis confirmed this result, with the median time to HF exacerbation being 37 days for spironolactone and 181 days for finerenone ($P < 0.001$).

This is illustrated by the KM curve in [Figure 2](#).

All-cause mortality

Mortality was significantly lower in the finerenone group (4.1%) compared to the spironolactone group (11.8%) with a risk difference of 7.7% (95% CI: 0.23–0.64) and an HR of 0.38 ($P < 0.001$). Kaplan–Meier (KM) analysis supported this result ($P < 0.001$).

This is illustrated by the KM curve in [Figure 3](#).

Acute kidney injury (AKI)

AKI occurred in 36% of patients on spironolactone and 28.3% on finerenone, with a risk difference of 7.7% (HR 0.81, 95% CI: 0.65–1.02). This difference was not statistically significant and may be explained by an underpowered sample size.

Hypokalaemia

Hypokalaemia occurred less frequently with finerenone compared to spironolactone (15.1% vs. 8.4%) with a risk difference of 6.7% (HR: 0.58 [95% CI: 0.40–0.85], $P = 0.001$).

This is illustrated by the KM curve later.

Hyperkalaemia

Hyperkalaemia rates were similar (16.1% with spironolactone vs. 13.8% with finerenone), with a risk difference of 2.3% (95% CI: 0.69–1.33), which was not significant ($P = 0.325$).

Sensitivity analysis

To further test the robustness of our findings, we performed quantitative bias analyses using E-values for our key outcomes (HF exacerbation and all-cause mortality, as well as secondary outcomes). E-values quantify the minimum strength of association that an unmeasured confounder would need to have with both treatment assignment and the outcome to fully explain away the observed association, beyond the measured covariates. For all-cause mortality the HR of 0.35 corresponds to an E-value of approximately 5.16 for the point estimate and 2.66 for the upper bound of the CI, indicating that only a relatively

Table 5 Baseline characteristics for patients with HFpEF with finerenone or spironolactone, stratified by DM, before and after propensity score matching

	Before matching			After matching		
	Finerenone	Spironolactone	P-value	Finerenone	Spironolactone	P-value
<i>n</i>	453	138 039		447	447	
Age at index	70.08 ± 11.24	68.9 ± 12.48	0.044	70.02 ± 11.26	69.62 ± 11.55	0.602
Sex						
Female	188 (41.5%)	70 883 (51.4%)	<0.001	187 (41.8%)	182 (40.7%)	0.734
Male	259 (57.2%)	62 971 (45.6%)	<0.001	254 (56.8%)	258 (57.7%)	0.787
Race						
White	243 (53.6%)	89 219 (64.6%)	<0.001	243 (54.4%)	236 (52.8%)	0.639
Black or African American	65 (14.3%)	30 345 (22.0%)	<0.001	65 (14.5%)	70 (15.7%)	0.640
Asian	83 (18.3%)	3324 (2.4%)	<0.001	77 (17.2%)	72 (16.1%)	0.654
American Indian or Alaska Native	10 (2.2%)	447 (0.3%)	<0.001	10 (2.2%)	10 (2.2%)	1.000
Native Hawaiian or Other Pacific Islander	14 (3.1%)	895 (0.6%)	<0.001	14 (3.1%)	10 (2.2%)	0.408
Other race	24 (5.3%)	3428 (2.5%)	<0.001	24 (5.4%)	25 (5.6%)	0.883
Unknown Race	22 (4.9%)	10 381 (7.5%)	0.032	22 (4.9%)	29 (6.5%)	0.313
Ethnicity						
Not Hispanic or Latino	366 (80.8%)	106 012 (76.8%)	0.044	360 (80.5%)	366 (81.9%)	0.607
Diagnoses						
Atrial fibrillation and flutter	177 (39.1%)	67 559 (48.9%)	<0.001	176 (39.4%)	172 (38.5%)	0.784
Chronic kidney disease (CKD)	391 (86.3%)	70 492 (51.1%)	<0.001	385 (86.1%)	390 (87.2%)	0.623
Diabetes mellitus	453 (100.0%)	138 039 (100.0%)	<0.001	447 (100.0%)	447 (100.0%)	0.664
Essential (primary) hypertension	416 (91.8%)	122 648 (88.9%)	0.044	411 (91.9%)	417 (93.3%)	0.443
Obstructive sleep apnea (adult) (pediatric)	201 (44.4%)	58 287 (42.2%)	0.356	200 (44.7%)	226 (50.6%)	0.082
Old myocardial infarction	99 (21.9%)	31 541 (22.8%)	0.615	99 (22.1%)	91 (20.4%)	0.513
Overweight and obesity	281 (62.0%)	86 136 (62.4%)	0.871	279 (62.4%)	297 (66.4%)	0.209
Medications						
Ace inhibitors	221 (48.8%)	75 074 (54.4%)	0.017	218 (48.8%)	226 (50.6%)	0.593
Angiotensin ii inhibitor	295 (65.1%)	66 875 (48.4%)	<0.001	290 (64.9%)	277 (62.0%)	0.367
Beta blockers/related	406 (89.6%)	123 170 (89.2%)	0.786	401 (89.7%)	401 (89.7%)	1.000
Glucagon-like peptide-1 (GLP-1) analogues	193 (42.6%)	15 478 (11.2%)	<0.001	187 (41.8%)	191 (42.7%)	0.787
Loop diuretics	377 (83.2%)	127 233 (92.2%)	<0.001	376 (84.1%)	385 (86.1%)	0.398
Sacubitril	37 (8.2%)	12 765 (9.2%)	0.428	37 (8.3%)	33 (7.4%)	0.618
Sodium-glucose co-transporter 2 (SGLT2) inhibitors	247 (54.5%)	27 484 (19.9%)	<0.001	241 (53.9%)	251 (56.2%)	0.501
Thiazides/related diuretics	252 (55.6%)	69 454 (50.3%)	0.024	250 (55.9%)	249 (55.7%)	0.946
Valsartan	97 (21.4%)	28 369 (20.6%)	0.651	97 (21.7%)	103 (23.0%)	0.630

Significant P-values in bold.

strong unmeasured confounder (with a risk ratio >5 with both treatment and outcome) could fully explain away the observed association, showing that the association between finerenone and all-cause mortality is highly robust to unmeasured confounding. A confounder would need to increase the likelihood of both receiving finerenone and of death by more than five-fold to negate the effect. Similarly, for HF exacerbation, a corresponding E-value point estimate of 3.77 based on odds ratio means that an unmeasured confounder would have to increase the risk by almost four times to nullify the observed results, further highlighting the robustness of our results. These sensitivity analyses, when performed for both primary and secondary outcomes, highlight the robustness of our observed findings across.

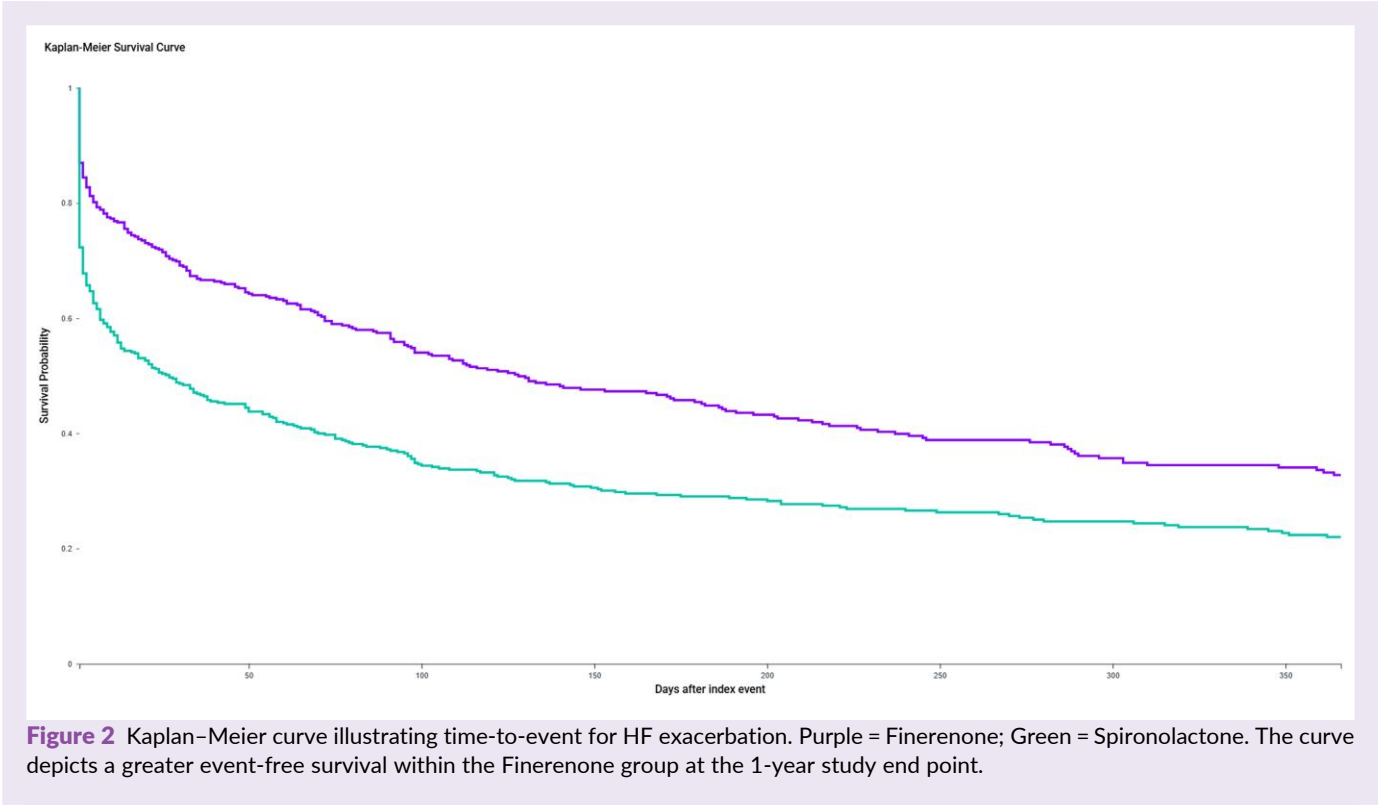
Discussion

Despite representing half of the HF population, HFpEF remains a formidable clinical challenge with limited pharmaceutical options.¹ In stark contrast to the compelling results purporting steroidal MRA use in HFrEF,^{6,7} mixed results were seen in the TOPCAT trial, the only large-scale trial investigating spironolactone in HFpEF.⁸ Subsequent post hoc analyses of the TOPCAT trial, along with large observational studies, demonstrated clinical benefit when using spironolactone in HFpEF in reducing hospitalizations or mortality.^{9,17,18} Herein, the emergence of finerenone as a selective, non-steroidal MRA offers a promising therapeutic option in HFpEF, as illustrated by the FINEARTS-HF trial, which divulged

Table 6 Outcomes between patients with HFpEF with finerenone or spironolactone, stratified by DM, after propensity score matching, with hazard ratios and 95% confidence intervals

	Finerenone	Spironolactone	HR (95% CI)	P-value
HF Exacerbation	253 (56.6%)	331 (74.0%)	0.59 (0.50–0.69)	<0.001
AKI	135 (30.2%)	171 (38.3%)	0.74 (0.59–0.93)	0.011
Renal Replacement therapy	10 (2.2%)	12 (2.7%)	0.78 (0.33–1.84)	0.666
AMI	37 (8.3%)	46 (10.3%)	0.83 (0.54–1.28)	0.300
Ischaemic stroke	29 (6.5%)	29 (6.5%)	1.04 (0.62–1.73)	1.000
All-cause mortality	17 (3.8%)	50 (11.2%)	0.35 (0.20–0.61)	<0.001
Sudden cardiac death	10 (2.2%)	10 (2.2%)	0.34 (0.07–1.70)	1.000
Hypokalaemia	38 (8.5%)	69 (15.4%)	0.55 (0.37–0.82)	0.001
Hyperkalaemia	65 (14.5%)	83 (18.6%)	0.78 (0.56–1.08)	0.105
Atrial fibrillation/flutter	138 (30.9%)	156 (34.9%)	0.87 (0.69–1.09)	0.200

Significant P-values in bold.



significant reductions in its primary outcome (a composite of total worsening HF events and deaths from cardiovascular causes).¹⁵ A recent 2024 meta-analysis by Jhund *et al.* reviewed all the data from the landmark MRA trials and concluded that all patients across the HF spectrum would benefit from MRAs, with non-steroidal MRAs specifically indicated for HFpEF mainly due to the data from the FINEARTS-HF trial.¹⁹ Yet, with no direct head-to-head comparison between spironolactone and finerenone, there is a critical gap in our understanding of optimal medical management within the HFpEF population. Our study, therefore, aims to offer real-world comparisons between the two MRAs

using a large, propensity-matched national cohort to help guide clinical decision-making and future prospective research.

In our cohort of 982 patients, finerenone was associated with significantly lower rates of HF exacerbation and mortality when compared to spironolactone. The risk of HF hospitalization was reduced by nearly half (HR = 0.63 [95% CI: 0.54–0.74], *P* < 0.001) in the finerenone group, with a further reduction in risk of all-cause mortality (HR = 0.38 [95% CI: 0.23–0.64], *P* < 0.001). Importantly, these associations for all-cause mortality and HF exacerbation rates remained statistically significant following stratification by CKD and DM. Such findings are

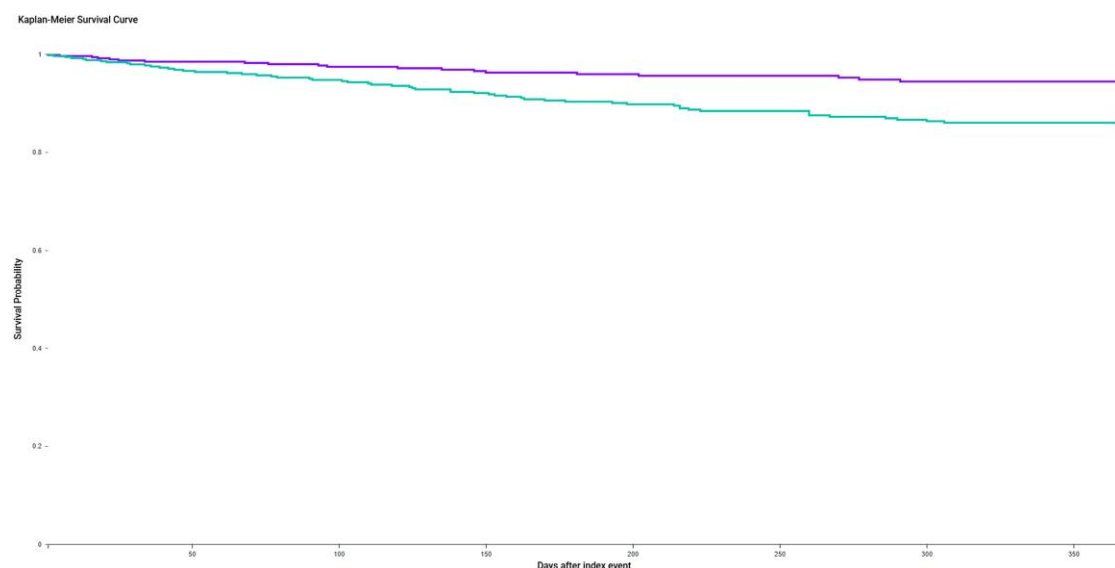


Figure 3 Kaplan–Meier curve illustrating time-to-event for mortality. Purple = Finerenone; Green = Spironolactone. The curve depicts a greater survival within the Finerenone group at the 1-year study end point.

compelling as they build upon the prior observations from the TOPCAT and FINEARTS-HF trials. The TOPCAT trial illustrated significant geographic heterogeneity, obscuring the true potential benefits of spironolactone,⁹ while the FINEARTS-HF trial failed to demonstrate a statistically significant reduction in cardiovascular mortality.¹⁵ By directly comparing finerenone and spironolactone, our study suggests that non-steroidal MRAs may be associated with reduction in both HF hospitalizations as well as improved overall survival benefit relative to spironolactone. Such differences may be driven by finerenone's improved receptor selectivity and favourable pharmacokinetics with a shorter half-life and no active metabolites, promoting tolerability and perhaps permitting sustained therapy without frequent dose adjustments or discontinuations due to side effects.^{11,20,21}

Our observational findings can be contextualized alongside the FINEARTS-HF trial, which evaluated finerenone in patients with HFpEF and HFmrEF and demonstrated reductions in a composite of HF outcomes, with some variation across clinical subgroups. Our real-world HFpEF population, which has a high burden of CKD and diabetes, demonstrated results that are generally directionally consistent with the trial's signal in terms of reduced HF events and improved survival. However, important differences in study design (randomized trial vs. observational database), inclusion criteria, and outcome definitions prevent direct comparison. We explicitly frame our data as hypothesis-generating real-world evidence that complements, but does not replace, the randomized evidence from FINEARTS-HF.

The findings from our study provide further insight into the safety profiles of these drugs, particularly with respect to renal injury and potassium deviations. Within the overall cohort, finerenone did not demonstrate significant reductions in the rates of AKI over the 1-year observation period (HR = 0.81 [95% CI: 0.65–1.02]). Our sample size may have been underpowered to detect a significant difference. However, statistically significant reductions in AKI were seen within our DM and CKD subgroup

analyses, a consistent finding seen within this patient population in past studies.^{13,22} It is worth noting that despite PSM, residual differences in underlying severity of renal disease and use of concomitant nephrotoxic medications may persist. Additionally, lack of access to newer therapies, insurance coverage, may contribute further to preferential use of spironolactone. We emphasize that these factors may confound the observed differences and that causality cannot be inferred. Additionally within the diabetes subgroup, patients treated with spironolactone appear to have a higher comorbidity burden and may represent a clinically sicker population than finerenone recipients. This can be explained by potential system-level and prescribing factors that could contribute to this pattern, including formulary restrictions and prior authorization requirements for finerenone, differences in insurance coverage, and variation by prescribing specialty (e.g. nephrology vs. cardiology vs. primary care). These factors may preferentially channel more severe or socioeconomically disadvantaged patients to generic spironolactone. Hence, even after matching, these baseline differences may potentially create a bias in comparison between both groups. We highlight this as a form of channelling bias and emphasize that subgroup results in the DM population should be viewed as exploratory.

We further observed a significant reduction in the risk of hypokalaemia within the finerenone group compared to spironolactone (HR = 0.58 [95% CI: 0.40–0.85], $P = 0.001$), indicating that finerenone may provide a more stable potassium balance compared to the traditional steroidal MRAs in HFpEF. Similarly, although not statistically significant, lower rates of hyperkalaemia were noted within the finerenone cohort (16.1% with spironolactone vs. 13.8% with finerenone; HR = 0.96 [95% CI: 0.69–1.33], $P = 0.325$). Prior studies including the FIDELIO-DKD and FIGARO-DKD trials that investigated finerenone vs. placebo within DM and CKD patients have consistently reported improvements in cardio-renal outcomes at the cost of higher incidences of hyperkalaemia^{23,24}; values that required permanent stoppage of finerenone remained low at 2.3% for FIDELIO-DKD and 1.2% for FIGARO-DKD. The FIDELITY

pooled analysis of the aforementioned trials further supported these findings.²⁵ Yet, when compared to spironolactone, our findings are in accordance with the expectations set by the Mineralocorticoid Receptor Antagonist Tolerability Study, where finerenone was associated with a lower incidence of hyperkalaemia.¹³ Similarly, in post-hoc analyses studying patients with CKD and treatment-resistant hypertension by Agarwal *et al.*, finerenone demonstrated significantly lower rates of hyperkalaemia compared to spironolactone alone and spironolactone with patiomer, a potassium binder (12% with finerenone and 3% with placebo, compared to 35% with spironolactone plus patiomer and 64% with spironolactone plus placebo). Consequently, treatment discontinuation due to hyperkalaemia was only 0.3% in the finerenone group vs. 7% for spironolactone + patiomer and 23% for spironolactone + placebo.²²

While our study corroborated the hypokalaemia findings from the FINEARTS-HF trial (i.e. lower hypokalaemia risk compared to placebo),¹⁵ several reasons may explain the statistically insignificant difference in hyperkalaemia rates observed in our population-level analyses. First, due to the observational nature of our study, our cohorts were likely underpowered to detect statistically significant differences in hyperkalaemia. Moreover, we were unable to control exact dosing regimens for either drug; such dosing nuances could narrow potential intrinsic advantages offered by finerenone. The prior evidence of finerenone's safety profile comes largely from lower-dose comparisons (10 mg),¹³ despite studied regimens ranging from 2.5 to 40 mg.^{14,15,23,24,26} As a result, higher finerenone dosing may produce potassium elevations more closely resembling spironolactone, especially when the latter is used conservatively. In addition, real-world practices such as increased frequency of monitoring or selective prescribing may reduce the hyperkalaemia risk of spironolactone. On the other hand, with finerenone being initially targeted for diabetic CKD and an option for patients prone to hyperkalaemia, it may be introduced to 'higher-risk' patients who previously had no MRA option. Lastly, while we did match numerous demographic factors, concurrent drugs, and comorbidities (as seen in Table 1), uncontrolled patient characteristics can further influence event rates. We were unable to determine the distribution of CKD severity or baseline eGFRs across our population. Uneven representation of CKD severity between groups would impact results with spironolactone is generally restricted to patients with reasonably preserved renal functions (with guidelines highlighting tolerability issues in patients with eGFR <30–45 mL/min or baseline serum potassium >4.5–5 mEq/L),^{4,27} while finerenone was studied in CKD down to eGFR 25.²³ Further, the use of potassium binders such as patiomer was not controlled for. In the AMBER trial, the addition of patiomer permitted ongoing spironolactone therapy with fewer hyperkalaemia-related withdrawals.²⁸ Our findings suggest that while finerenone may have a mechanistic advantage and applications in high-risk populations, both MRAs would require regular potassium surveillance and individualized treatment to maximize safety.

The results of this study have important clinical implications for the management of HFpEF, a condition that historically lacked efficacious therapies to improve prognosis, with current treatment largely revolving around symptom management and comorbidity control. Our findings suggest that finerenone could emerge as a valuable therapeutic option in this space, building upon the results of the FINEARTS-HF trial. However, our observational study is subject to residual confounding and selection bias, and these findings should be considered hypothesis-generating. Nonetheless, our findings illustrate a need and a strong rationale for direct large-scale, head-to-head studies to clearly delineate the risk-benefit considerations between the two agents. Confirmation in randomized clinical trials and

prospective studies is required before any changes to clinical practice can be recommended based on our results. If confirmed in prospective trials, finerenone may supersede spironolactone as the preferred MRA for HFpEF, offering greater reductions in hospitalizations and a mortality benefit along with a favourable renal risk profile, lower risk of electrolyte derangements, and AKI. This would mark a significant advancement in HFpEF care, supporting a more prominent role for MRAs and effectively expanding the therapeutic toolkit for a disease that, until recently, had a paucity of evidence-based treatments.

Implications for real world use

Real-world data for evaluation of the implementation of GDMT in clinical practice is increasingly important, particularly due to the fact that randomized control trials typically involve selected patient populations and close monitoring protocols that often do not reflect routine, day-to-day clinical care. Prior registry-based studies have demonstrated that individuals enrolled in quality registries tend to be more adherent to guideline-recommended therapies, and, as a result, experience improved outcomes, though this may limit generalizability to broader real-world patient populations.²⁹ The EVOLUTION HF study conducted by Savarese *et al.*, revealed significant variability in initiation, up-titration, and discontinuation of GDMT patterns between evidence-based recommendations and real-world practice, particularly with novel therapies.³⁰ The CHAMP-HF registry revealed that fewer than 25% of eligible HFpEF patients were prescribed all three of the main classes of GDMT therapy, with fewer than 1% achieving targeted doses of this medication regimen, underscoring the rather large gap in clinical practice.³¹ Our findings, via the TrinetX database, reflect real-world treatment patterns among selected HFpEF patients receiving steroidal and non-steroidal MRAs and provide complementary insights into the real-world uptake and outcomes of contemporary HF therapies outside of controlled environments ubiquitous in clinical trials.

Strengths and limitations

Our study does offer unique strengths. By leveraging a large, multi-centre electronic health record database (TriNetX), we assembled a diverse HFpEF cohort. We adopted a propensity-score matching approach, which balanced numerous baseline characteristics between the two groups and mimicked a quasi-randomized study design, in turn mitigating some selection biases and measured confounding typically associated with observational studies. Through additional stratified sub-analyses of DM and CKD populations, we aimed to further minimize potential selection bias, as finerenone is often prescribed for indications including DM and/or CKD. Our enrolment of heterogeneous populations from varied healthcare settings can better reflect typical outcomes in everyday settings, promoting stronger external validity relative to RCTs, which often are restricted to more selective patient populations. As an example, we were able to enrol a diverse patient population, yielding a cohort more representative of the general US population and directly addressing a limitation reported within the FINEARTS-HF trial, where they had few Black patients.¹⁵ With our focus on concrete clinical outcomes, such as hospitalization for HF and all-cause mortality, we are able to align with the contemporary standards for HFpEF research and enhance the pragmatic relevance of our findings.

Nonetheless, several limitations afflict our present study. First, this is a retrospective observational study and hence cannot prove causation nor can it fully eliminate bias and confounding, despite robust propensity matching and covariate adjustment. As previously illustrated with regard to hyperkalaemia, unmeasured

confounders can influence results. We attempted to account for known risk factors, but differences in aspects like HF severity (e.g. NYHA scoring or diastolic function) or CKD severity (e.g. eGFR levels or need for renal replacement therapy) were not captured in structured fields within TriNetX, and hence unable to be included in our study. As a result, measures of symptom burden and HF severity may contribute to residual confounding, which cannot be addressed through adjustment or matching. Second, this confounding bias is further accentuated by the differences in sample size within the finerenone and spironolactone cohorts, with the finerenone cohort being notably smaller than the latter. Similarly, while we controlled commonly used medications in HF, other adjunct medications like patiromer, which can exorbitantly impact the electrolyte profile, were not accounted for. Moreover, the identification of HFpEF was based on ICD-10 code I50.3 recorded in the electronic health record; echocardiographic parameters (e.g. left ventricular ejection fraction, diastolic indices) were not consistently available. We acknowledge the potential for misclassification of HF phenotype: some patients with HF with improved ejection fraction or mid-range ejection fraction may have been coded as HFpEF, and conversely some true HFpEF cases may have been missed. However, the use of the ICD-10 code is likely to be specific for patients who had sufficient evidence to be labelled as HFpEF by healthcare providers. Hence, our findings from this study apply to this patient population coded as HFpEF and may not generalize to all patients with suspected HFpEF in the real world. The finerenone group contained substantially less subjects compared to the spironolactone population, which may cause channelling and formula effects. Hence, residual confounding by unmeasured variables cannot be excluded. Patients initiated on finerenone often had more comorbidities, such as diabetes and CKD, compared to spironolactone in the pre-matching cohort, despite matching and covariate adjustment. We did not have granular data on medication dose titration or medication adherence. It is unknown if both groups were on their maximally tolerated or optimal dosing, thus hindering comparative efficacy. Treatment groups were defined based on the index MRA and patients were followed in an intention-to-treat-like manner. On the TriNetX database, exact discontinuation dates, or reasons for stopping or switching therapy, such as due to hyperkalaemia cannot be determined. Hence, we are unable to determine with certainty whether patients had received finerenone as the initial medication of choice or if there were prior MRAs used before initiation of finerenone. Additionally, survivor bias and prevalent user bias in the finerenone group cannot be excluded, as these patients may have survived long enough to be initiated on the medication. Time zero was defined as the first prescription date, and only post-index events were counted; therefore, we cannot fully exclude immortal time-related biases. It is worth noting that patients within the finerenone cohort may have been managed in more specialized settings with better access to specialists and care, highlighting potential selection bias. These phenomena may partially explain the observed differences in outcomes and our findings cannot be interpreted as proof of causal benefit. Hence, our findings are hypothesis-generating in a coded HFpEF population and may not naturally generalize to all HFpEF populations. A prospective, head-to-head randomized controlled trial of finerenone vs. spironolactone in HFpEF would be the definitive way to confirm our observations.

Conclusion

HFpEF continues to portend significant morbidity and mortality within the US and globally. While there is increasing fervour surrounding the use of the non-steroidal MRA finerenone, data

regarding the comparative efficacy and safety profile of the non-steroidal MRA finerenone and steroidal MRA spironolactone remain sparse. Our study demonstrates that finerenone use was associated with a comparative therapeutic advantage with a better safety profile in this population. These observational findings are hypothesis-generating and require confirmation via a prospective, head-to-head randomized controlled trial of finerenone vs. spironolactone in HFpEF to further inform this debate.

Supplementary material

Supplementary material is available at *European Heart Journal—Quality of Care* and *Clinical Outcomes* online.

Author contributions

Talal Almas (Conceptualization [lead]; Data curation [lead]; Formal analysis [lead]; Investigation [lead]; Project administration [lead]; Writing—original draft [lead]; Writing—review & editing [lead]), Daniel I Simon, Claire Sullivan, Ravi Ramani, Michael P Zacharias, Ali Alansari (Investigation [supporting]; Writing—original draft [equal]; Writing—review & editing [supporting]), Mariam Nadeem Rana, Abdullah Almukhaizeem (Supervision [supporting]; Validation [supporting]; Writing—original draft [supporting]; Writing—review & editing [supporting]), Khaled AlMojil (Project administration [supporting]; Resources [supporting]; Supervision [supporting]; Validation [supporting]; Writing—review & editing [supporting]), Abdullah W AlHamlan (Resources [supporting]; Validation [supporting]; Visualization [supporting]; Writing—review & editing [supporting]), Jasem Burahmah (Project administration [supporting]; Validation [supporting]; Writing—original draft [supporting]; Writing—review & editing [supporting]), Joseph A Reed, Mahmoud Ismayl, Amal Naveed Qureshi (Methodology [supporting]; Visualization [supporting]; Writing—original draft [supporting]; Writing—review & editing [supporting]), Absam Akbar, Kannu Bansal, Ahmed Al-Hindawi, Mohamed Shelig, Nihar R Desai, and Thomas Joseph O'Neill

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Data availability

The data underlying this article are available in the article and its online [supplementary material](#).

The data underlying this article were provided by TriNetX under licence. Data will be shared on request to the corresponding author with permission of TriNetX.

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